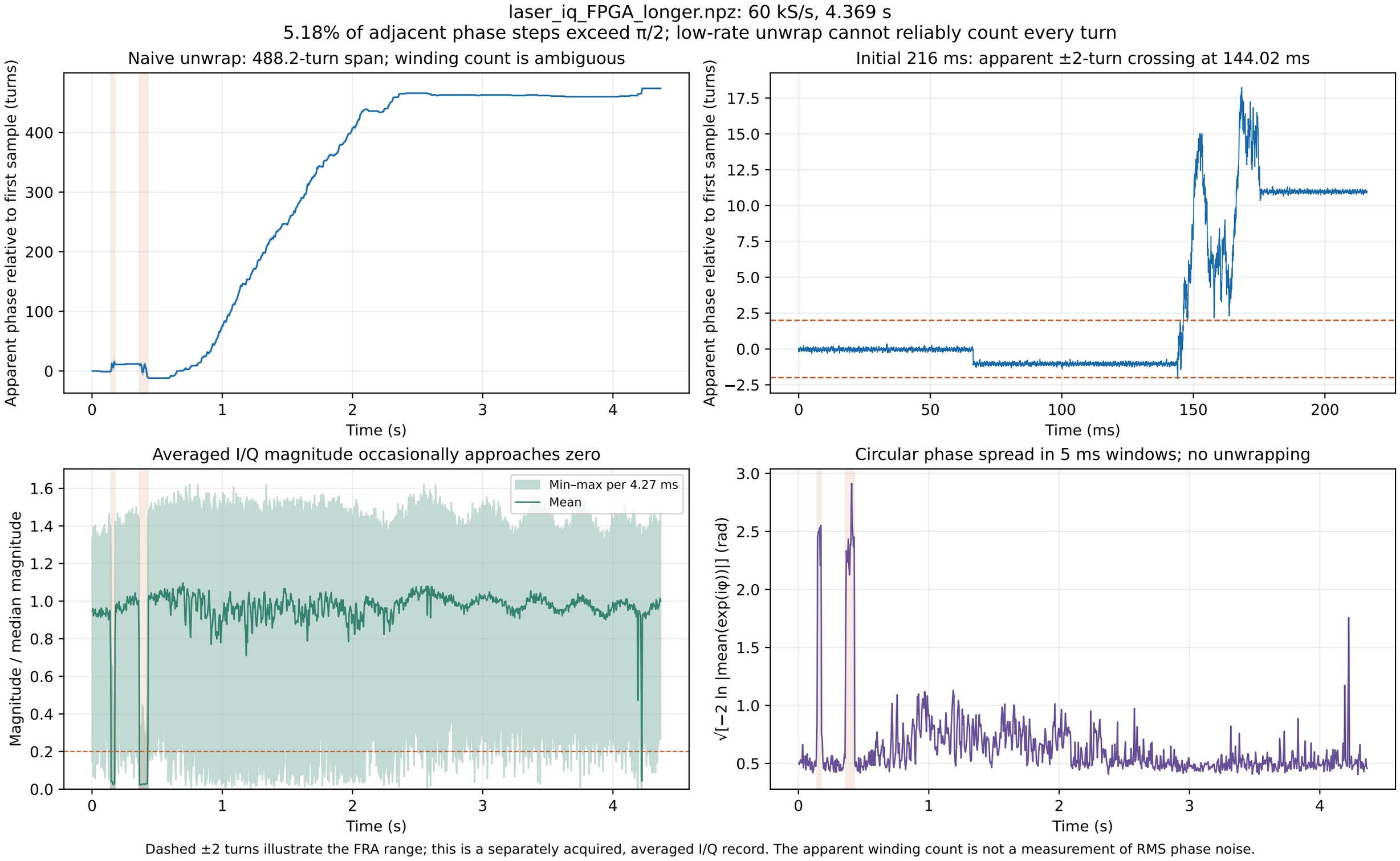

laser_iq_FPGA_longer.npz: 60 kS/s, 4.369 s
5.18% of adjacent phase steps exceed π/2; low-rate unwrap cannot reliably count every turn
Naive unwrap: 488.2-turn span; winding count is ambiguous
Apparent phase relative to first sample (turns)
Time (s)
Initial 216 ms: apparent ±2-turn crossing at 144.02 ms
Apparent phase relative to first sample (turns)
Time (ms)
Averaged I/Q magnitude occasionally approaches zero
Min–max per 4.27 ms
Mean
Magnitude / median magnitude
Time (s)
Circular phase spread in 5 ms windows; no unwrapping
√[−2 ln |mean(exp(iφ))|] (rad)
Time (s)
Dashed ±2 turns illustrate the FRA range; this is a separately acquired, averaged I/Q record. The apparent winding count is not a measurement of RMS phase noise.